02LQD Specification and Simulation of Digital Systems

2023/09/21

1. Combinational circuit design (2 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSEIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A **median filter** removes **lone 1s** in the input stream by changing each **lone 1** to a 0 on the output. A lone 1 is a 1 « sandwiched » between two 0s.

Example (lone 1s in boldface)
input stream: 10110**1**01110**1**0**1**01
output stream: 10110**0**01110**0**0**0**01

Note that the output stream is the (modified) input stream 2 two clock ticks later.

Design a Moore FSM with one serial input X and one serial output Z compliant with the specifications. Reset is asynchronous.
Design a **Moore** FSM with the above specified inputs, outputs and behavior:

- Draw the state diagram (**4 points**)
- Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behaviour as the 2-process description (**4 points**).
- Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

4. FSM-D design (16 points)

Tribonacci numbers are recursively defined as follows:

$TRIB_n = TRIB_{n-3} + TRIB_{n-2} + TRIB_{n-1}$ $\quad n \geq 3$
$TRIB_0 = 0$
$TRIB_1 = 1$
$TRIB_2 = 1$

The first 10 Tribonacci numbers are thus: 0 1 1 2 4 7 13 24 44 81.

Design a FSM-D to **iteratively** compute Tribonacci numbers of order `n`, where `n` is on 4 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output is `P`. Define the appropriate size for `P`. The output `P` makes sense only when `ready` is 1.

Steps:

1. Define a proper size for the output `P` (**2 points**)
2. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. Design a FSM-D:
   - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**).
   - create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

Exam rules:

• upload on Portale della didattica in section Elaborati a .zip archive including:
  ○ running VHDL code and testbench
  ○ short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
no later than Sunday September 24 11.59 pm
• format of .zip archive name surname_ID (example Rossi_244123)
• remember: no upload ⇒ no correction ⇒ no oral exam
• once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.